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Paleomagnetic evidence for Jurassic deformation of the McCoy Mountains Formation, southeastern California and southwestern Arizona

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The Mesozoic McCoy Mountains Formation is a 7.3-km-thick deformed clastic sequence exposed in six mountain ranges in southeastern California and southwestern Arizona. Interbedded with Jurassic volcanic rocks at its base, the McCoy Mountains Formation had been assigned a Cretaceous age based upon fossil angiosperm wood found in the upper third of the section. Characteristic natural remanent magnetism (NRM) from 145 oriented samples from 18 sites within the sedimentary terrane yield an in situ mean direction: $I = 20.6°$, $D = 335.1°$, $\alpha_{95} = 7.7°$ (uncorrected for structural tilting). Opaque mineralogy and a failed fold test indicate that the NRM is a chemical remanence acquired post-folding. The paleomagnetic pole position calculated from the in situ mean direction falls adjacent to poles from the Summerville Formation and Canelo Hills Volcanics. We interpret these data to indicate that deformation, mild metamorphism, and resultant magnetization of the McCoy Mountains Formation occurred during Jurassic time. It is suggested that the McCoy Mountains Formation and underlying Jurassic volcanics were deposited adjacent to, and then deformed between, the North American craton and an outlying allochthonous terrane during Jurassic time.

1. Introduction

Stratigraphy and isotopic dating have provided only broad age constraints on the McCoy Mountains Formation, a Mesozoic sedimentary terrane exposed in six mountain ranges in southeastern California and southwestern Arizona. The sedimentary rocks of the terrane are exceedingly thick, lithologically monotonous, and very different from the classic Mesozoic sequence exposed on the Colorado Plateau, only 250 km to the northeast. Furthermore, the McCoy Mountains Formation has not been included in any regional tectonic syntheses of southwestern North America. In an effort to provide better age constraints, a paleomagnetic study was undertaken on the McCoy Mountains Formation. We present here the results of this study.

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2. Geology of the McCoy Mountains Formation

2.1. Sedimentology and stratigraphy

The McCoy Mountains Formation is exposed in the McCoy, Palen, and Coxcomb Mountains, in Riverside County, California, and in the Dome Rock, southwestern Plomosa and New Water Mountains and Livingston Hills in Yuma County, Arizona (Fig. 1). Geographically, the known exposures of McCoy Mountains Formation define a west-northwest-trending band 140 km long by 25 km wide. The term McCoy Mountains Formation as used here includes (1) the McCoy Mountains

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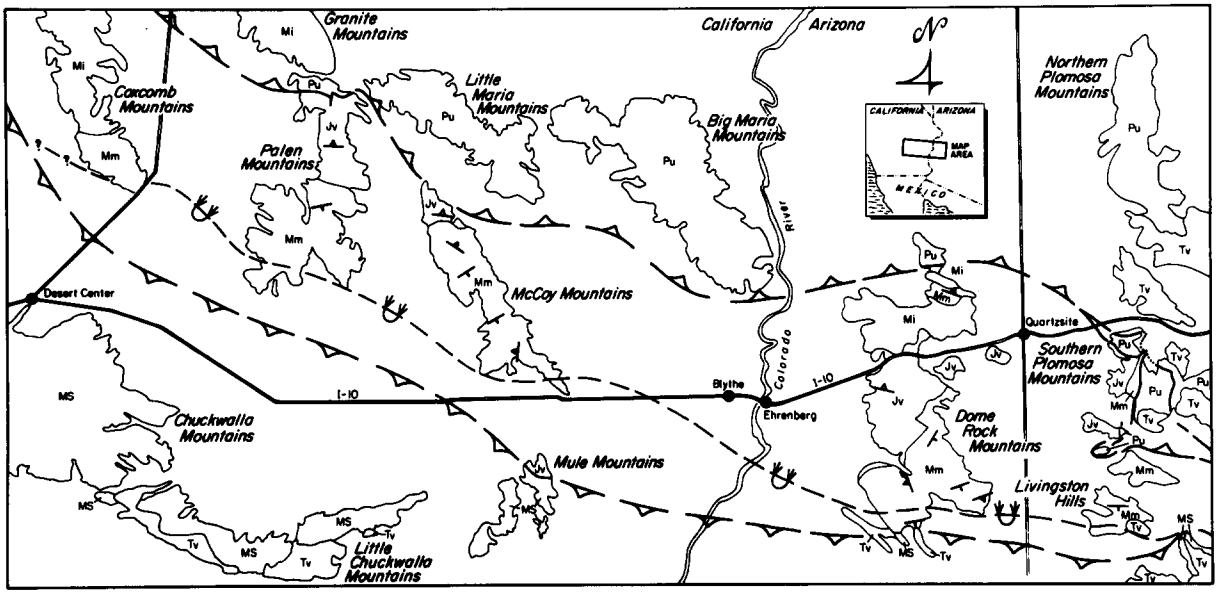


Fig. 1. Tectonic setting of the McCoy Mountains Formation in southwestern California and southwestern Arizona. Map legend: *Tv* = Tertiary volcanic rocks, *Mi* = Mesozoic intrusive rocks, *Mm* = McCoy Mountains Formation, *Jv* = Jurassic volcanic rocks, *Pu* = Precambrian, Paleozoic, and Mesozoic rocks of North America, *MS* = Mojave-Sonora composite terrane.

Formation of W.J. Miller [1], Pelka [2], and Greene [3], (2) the Livingston Hills Formation of F.K. Miller [4,5], Crowl [6], Marshak [7,8] and Harding [9,10] and (3) the continental redbed deposits of F.K. Miller [4,5] and Robison [11,12]. The stratigraphic equivalence of these formational units is well established [13].

A heterogeneous volcanic sequence which has yielded a late Triassic/early Jurassic U-Pb age date [5] and an apparent 176 Ma (K-Ar, plagioclase) age [2] underlies and is interbedded at its top with basal McCoy Mountains Formation. The stratigraphic top of the McCoy Mountains Formation is nowhere exposed [13].

The McCoy Mountains Formation consists of sandstone, conglomerate, siltstone, and mudstone and has been divided into six members, each containing internally consistent petrology and stratigraphy [13]. Although a complete, unfaulted sequence is nowhere exposed, the McCoy Mountains Formation is 7.3 km thick in its type section in the McCoy Mountains, California [13,14]. Detailed petrographic analysis of the sandstone, metamorphosed to greenschist facies, suggests that the McCoy Mountains Formation was derived from a geologically heterogeneous source area containing plutonic, volcanic, and sedimentary rocks. The proportions of these source constituents evolved with time [13]. Paleocurrent indicators consisting of high-angle cross-strata and asymmetric ripple marks distributed vertically and laterally throughout the exposures of the McCoy Mountains Formation indicate south-flowing paleocurrents [13]. Sparse sedimentary structures including cross-strata, ripple marks, channels, wedge-shaped depositional units, and fining upwards sequences suggest a fluvial environment of deposition [13]. In addition, poorly preserved fossil wood, identified as angiosperm (?) was found in several localities within the upper one-third of the McCoy Mountains Formation [2,13,15].

2.2. Structure

The McCoy Mountains Formation is homoclinal and south- or southeast-dipping and its uppermost strata are deformed into a north-verging, overturned synclinal fold [13] (Fig. 2). A penetra-

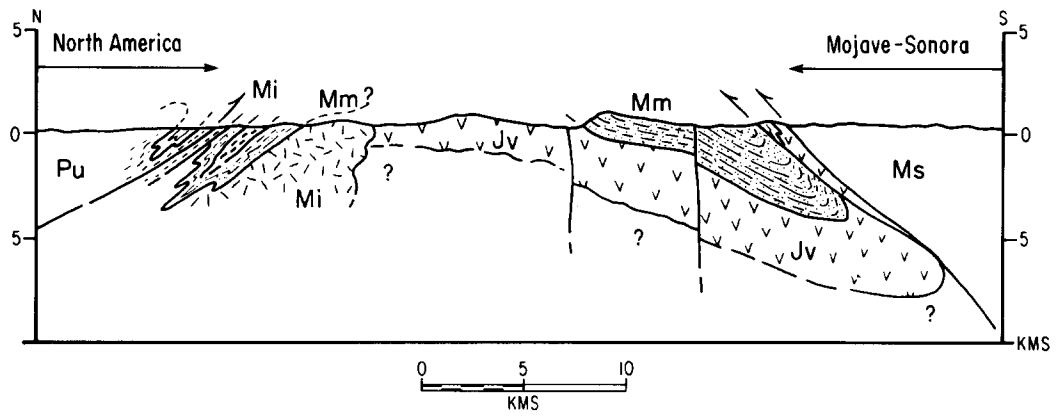


Fig. 2. Structure section through the Dome Rock Mountains, Arizona.

tive, south-dipping mineral cleavage sympathetic to the overturned syncline is present in the southern exposures of the McCoy Mountains Formation. The northern exposures of McCoy Mountains Formation and underlying Jurassic volcanic rocks contain a similarly penetrative, north-dipping mineral cleavage [13]. The two cleavages are never seen to cut one another; instead, one cleavage passes into the other within a 1-km-wide transition zone by a gradual change in structural attitude [13]. Throughout the change in cleavage orientation bedding remains south- or southeast-dipping. Attendant to the mineral cleavage is a mineral lineation which plunges moderately to the northwest where the cleavage dips north and moderately to the southeast where the cleavage dips south. The mineral lineation is not as penetrative as the mineral cleavage and is best observed in and near the cleavage transition zone.

Sympathetic to the north- and south-dipping mineral cleavages and overturning of folds are two thrust faults which form northern and southern limits of exposure of the McCoy Mountains Formation and underlying Jurassic volcanic sequence (Fig. 2). The northern bounding thrust fault, sympathetic to the north-dipping cleavage, verges south and places Precambrian, Paleozoic and lower Mesozoic rocks of the North American craton onto the Jurassic volcanic sequence and overlying McCoy Mountains Formation. The southern bounding thrust fault, sympathetic to the south-dipping cleavage, places crystalline rocks of the combined San Gabriel–Joshua Tree terranes of Powell [16] onto overturned McCoy Mountains Formation and underlying Jurassic volcanic rocks [13,17]. Thus the McCoy Mountains Formation and underlying Jurassic volcanic rocks appear to have been overridden by two thrust plates with opposing vergence, squeezed between rocks belonging to the North American craton and the allochthonous San Gabriel–Joshua Tree terrane.

Foliated McCoy Mountains Formation is intruded by the undeformed Coxcomb Pluton in the southern Coxcomb Mountains. The Coxcomb Pluton has yielded apparent ages of 71–65 Ma (K-Ar, biotite [2, 18,19]) thus providing a minimum upper Cretaceous age of deformation of the McCoy Mountains Formation. Wholesale resetting of older K-Ar systems to upper Cretaceous ages is well documented within the Mojave Desert [20] and the apparent upper Cretaceous ages from the Coxcomb Pluton should be considered to be preliminary, minimum ages. Calzia [21] reports a preliminary 3-point whole-rock Rb/Sr isochron from one phase of the Coxcomb Pluton yielding an apparent age of 145.4 Ma. If these data are supported by further study, they will substantiate the paleomagnetic data presented here.

Flat-lying middle Tertiary volcanic rocks rest with angular unconformity on overturned McCoy Mountains Formation in the Livingston Hills, Arizona [4,13]. Thus, there is no evidence that

tilting of the McCoy Mountains Formation occurred during the Basin and Range Orogeny.

2.3. Age constraints

The foliated McCoy Mountains Formation is interbedded at its base with Jurassic volcanic rocks and is intruded by an upper Cretaceous (?) pluton. Fossil wood, possibly angiosperm, has been collected from several localities within the upper one-third of the McCoy Mountains Formation but is generally in a poor state of preservation. This fossil wood has been used to place a late Cretaceous or Paleocene age [2,5,11,12,15] on the McCoy Mountains Formation although Jurassic volcanic rocks are interbedded at its base. New collections of the wood, recently examined, have been re-identified as angiosperm. However, it is understood that there exists considerable disagreement regarding the geochronologic limits of angiosperm wood. The dissent concerns: (1) disagreement as to the age of the oldest angiosperm pollen, (2) discrepancies between the age of the oldest angiosperm wood and angiosperm pollen, (3) lack of fossil angiosperm wood available for examination, and (4) the sudden appearance of angiosperm wood in the fossil record without evolutionary predecessors [22–26]. Therefore, even if the fossil wood from the McCoy Mountains Formation is angiosperm, the traditional Barremian [26] and younger age range for angiosperm wood may not hold. In conclusion, it is believed that the age constraints for the McCoy Mountains Formation are not confined to the late Cretaceous and Paleocene, as is commonly thought, but may extend throughout most of the Mesozoic.

3. Paleomagnetism

3.1. Procedures and techniques

In an effort to constrain further the age of the sedimentary terrane, a preliminary paleomagnetic study was done on the rocks of the McCoy Mountains Formation in the McCoy Mountains [27]. The study was then enlarged to include data gathered from the Palen and Dome Rock Mountains, with all data and results presented here.

A total of 79 oriented cores was collected from 11 sites in the McCoy Mountains and 66 oriented cores from 7 sites in the Dome Rock Mountains. The sites in the McCoy Mountains were evenly spaced within the lowest 1300 m of section; in the Dome Rock Mountains the lowest 1200 m of section were evenly sampled. Aerial distance between collecting localities in the McCoy and Dome Rock Mountains is about 60 m. Cores were taken using the standard coring techniques of Doell and Cox [28]. A Brunton compass was used to orient the cores. Paleomagnetic sites were restricted to purple and gray siltstone units. Twenty additional sites in the Dome Rock Mountains, six additional sites in the McCoy Mountains, and ten sites in the Palen Mountains were also cored. The data from these sites were rejected due to failure to isolate a characteristic natural remanent magnetization (NRM) direction for any of the sites.

NRM was measured on a cryogenic magnetometer. Alternating-field (AF) demagnetization was performed with a single-axis demagnetizer and thermal demagnetizations were done in a mu-metal shielded furnace. All thermal demagnetizations were done in air. The residual magnetic field in the cooling chamber was less than 10 gammas (1 gamma = 1 nanotesla).

3.2. Stability of NRM

NRM of all samples were measured before progressive demagnetization experiments. Intensities of NRM ranged from 10^{-4} to 10^{-1} A/m. Representative samples from each site were subjected to progressive AF and thermal demagnetization. Two different responses to demagnetization were observed, leading to rejection of many of the sites.

In the sites in which a characteristic NRM was isolated, progressive AF demagnetization up to 60 mT peak alternating field did not affect the NRM direction or intensity. Samples from these sites were then subjected to progressive thermal demagnetization. The results of the thermal demagnetization showed that the NRM contained almost no low blocking temperature secondary components of magnetization. Only a small fraction of the blocking temperature spectrum was below 580°C

(Curie temperature of magnetite), and many samples retained more than 50% of their original NRM after thermal demagnetization to 640°C. Fig. 3 is a vector demagnetization diagram of response to thermal demagnetization of a typical sample from a site in which a characteristic NRM was isolated. The high blocking temperatures and high coercivity indicate that hematite is the dominant ferromagnetic mineral present. Fig. 4A, B shows typical within-site grouping of sample NRM directions after thermal demagnetization.

The sites which were rejected showed a different response to the demagnetization techniques. Samples from the discarded sites in the Dome Rock and McCoy Mountains displayed anomalously weak NRM (10^{-4} A/m) intensities and scattered NRM directions that changed erratically during progressive AF or thermal demagnetization. The discarded sites in the Palen Mountains showed NRM directions parallel to present field directions. The NRM decreased in intensity when subjected to both AF and thermal demagnetization but did not change direction or improve in clustering. This response suggested that both hematite and magnetite are present as ferromagnetic minerals. The sites were discarded because a characteristic NRM could not be isolated. The

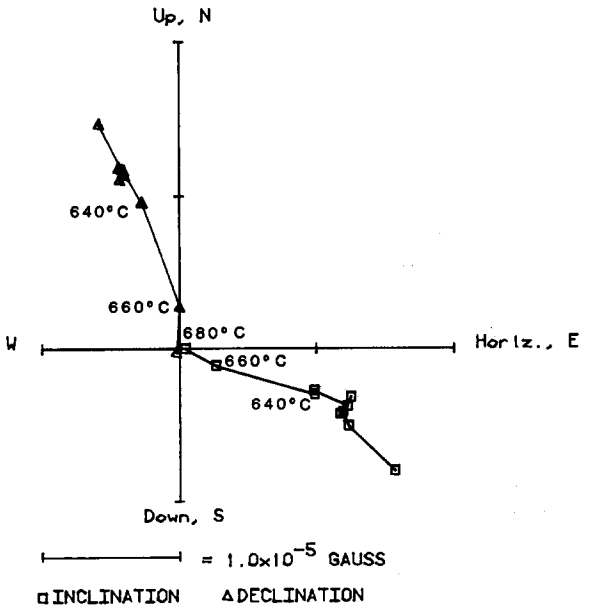


Fig. 3. Vector demagnetization diagram of response to thermal demagnetization of a sample from site (MC034) from which a characteristic NRM was isolated. Labels indicate temperatures of thermal demagnetization.

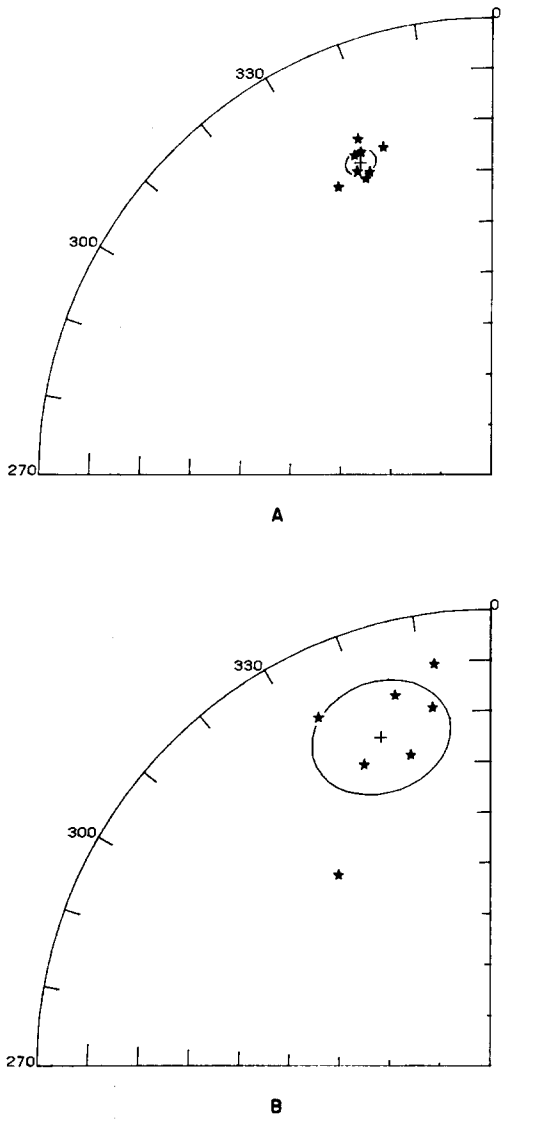


Fig. 4. A. Characteristic NRM directions for samples from MC002. B. Characteristic NRM directions for samples from MC006. Projections are equal area stereographic and all vectors are in the lower hemisphere. Oval of 95% confidence is shown surrounding site mean direction.

complex opaque mineralogy of these sites is discussed below.

Although no significant secondary components were present in the progressive demagnetization experiments from the sites which were finally included in this study, all samples were thermally demagnetized to 640°C. For most sites, no significant change of site mean direction resulted from this treatment. However, within-site grouping of NRM vectors was significantly improved by the thermal demagnetization treatment.

3.3. Opaque mineralogy

Microscopic examination of polished thin sections of representative samples from sites in which a characteristic NRM was isolated indicates that specular hematite is the only opaque mineral present in abundance. Textural evidence further indicates that the specular hematite was formed by post-depositional precipitation. Grains of specular hematite invade solution pits in adjacent silica grains and often have delicate growths which could not have been transported. No rounded specular hematite grains were observed, and almost all grains had sharp corners resulting from having filled angular voids between detrital grains. These observations are fully consistent with the paleomagnetic observations of very high coercivity with most blocking temperatures exceeding the 580°C Curie temperature of magnetite. The textural evidence indicating post-depositional precipitation of the specular hematite is also consistent with the results of the fold test described below. The opaque mineralogy thus indicates that the NRM is a chemical remanence acquired during in situ precipitation of specular hematite.

Samples from the sites which were discarded due to the inability to isolate a characteristic NRM have a different opaque mineralogy. Unlike the previously described samples, these samples are rich in detrital magnetite grains which are replaced in the amount of 70–80% by specular hematite. Specular hematite is also present in the material interstitial to the detrital grains. Although we are not certain of the explanation, it is evident that the presence of detrital magnetite grains (now largely oxidized to hematite) prevents the isolation of a characteristic NRM. The petrologic and composition differences between the two types of sites provides additional justification for exclusion of sites from which a characteristic NRM was not successfully isolated.

3.4. Fold test

The fold test of Graham [29] was applied to determine whether the NRM of the rocks was acquired before or after formation of structural tilting between sites. If the NRM was acquired *before* the relative structure between the sites was formed, site mean directions should converge when sites are corrected for structural tilting and show poorer grouping before structural correction. If the NRM was acquired *after* the relative structures between the sites were formed, site mean directions should cluster when not corrected for structural tilting and scatter upon correcting the directions for structural tilting.

Sites drilled in the McCoy Mountains Formation come from both north- and south-dipping cleavage domains and in both lower and upper limbs of the overturned syncline deforming the top of the McCoy Mountains Formation. The sites in the overturned fold limb display extremely weak ($< 10^{-4}$ A/m) and unstable NRM directions and were discarded. However, bedding attitudes in the lower, upright fold limb display sufficient variation to allow easy application of a fold test. One site (MC047) was drilled in locally transposed bedding in the domain of south-dipping cleavage.

The formation mean direction was calculated by computing the mean direction of the 17 site mean directions both before (uncorrected) and after (corrected) restoring the beds to horizontal at each site. These site mean directions are illustrated in Fig. 5A, B. The best estimate of the Fisher [30] precision parameter (k) was calculated for both uncorrected and corrected formation mean directions. The statistical significance of the fold test was evaluated by the method of Watson [31], using the tabulated F-ratio tables of McElhinny [32]. The k_1 (uncorrected) to k_2 (corrected) ratio is 4.21. A ratio of k_1/k_2 greater than 2.32 for a test with 17 directions indicates that the fold test is significant at the 99% confidence level. This test

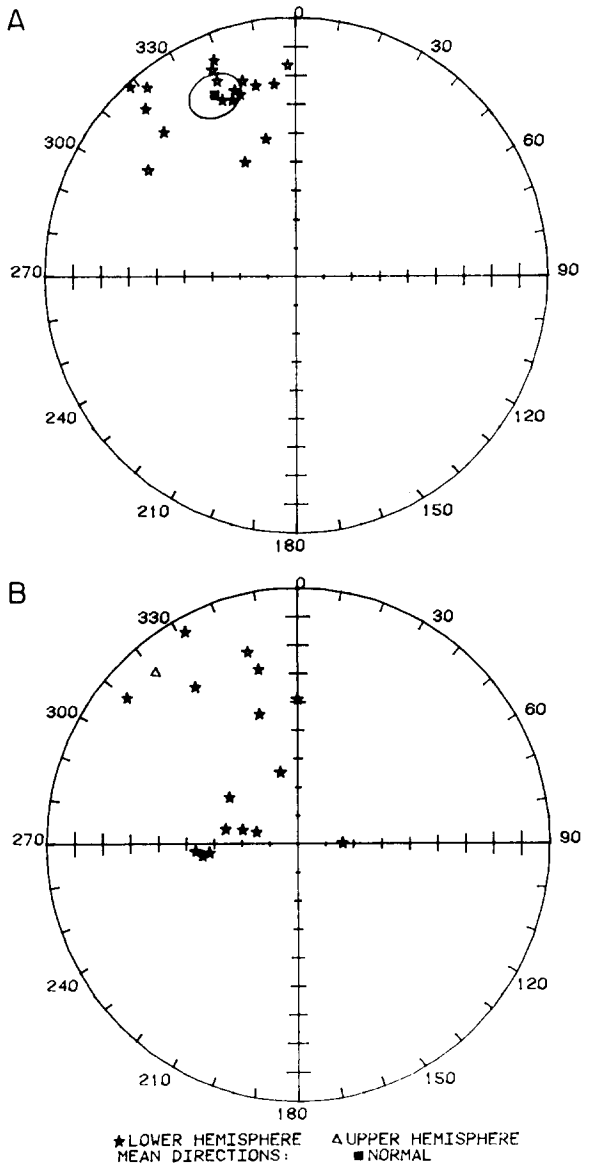


Fig. 5. A. Site mean and formation mean directions of characteristic NRM, not corrected for structural tilting. B. Site mean and formation mean directions of characteristic NRM, corrected for structural tilting. Projections are equal area stereographic. Oval of 95% confidence is shown surrounding mean direction.

indicates that the magnetization of the McCoy Mountains Formation was acquired *after* the deformation which produced the relative structural tilting between the sites. The paleomagnetic data thus indicate that the magnetization was acquired after deposition and deformation of the McCoy Mountains Formation and was produced by precipitation of hematite in response to a mild metamorphism. The characteristic NRM is thus a post-deformation chemical remanent magnetism.

3.5. Results

Both the opaque mineralogy and the fold test indicate that the magnetization of the McCoy Mountains Formation was acquired in response to a mild in situ metamorphism following deposition and deformation. As a result the site mean directions in Table 1 are in situ directions which have not been corrected for structural tilting. The formation mean direction and associated statistical parameters in Table 2 are obtained from the in situ site mean directions in Table 1. These in situ site mean directions are illustrated in Fig. 5A.

The mean direction of magnetization ($I = 20.6°$, $D = 335.1°$, $\alpha_{95} = 7.7°$) of the McCoy Mountains Formation from exposures in the McCoy and Dome Rock Mountains yields a paleomagnetic pole at $\lambda_p = 57.7°$N, $\phi_p = 116.2°$E with $dp = 4.2°$

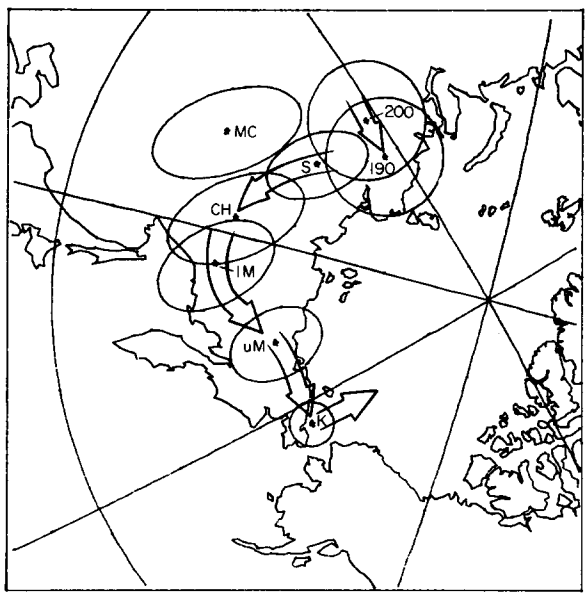


Fig. 6. Late Triassic through Cretaceous apparent polar wander path for North America. Pole data are listed in Table 3.

TABLE 1

Site mean directions of characteristic NRM

Site	$I(°)$	$D(°)$	$\alpha_{95}(°)$	k	N	R	Attitude of bedding(°)	Attitude of cleavage(°)
MC001	24.80	339.85	3.5	707.2	4	4.00	285, 75	90, 50
MC002	23.31	336.98	2.5	484.1	8	7.99	285, 75	90, 50
MC003	23.67	342.59	1.6	1165.5	8	7.99	285, 75	90, 50
MC004	22.68	353.56	7.1	306.6	3	2.99	280, 65	90, 45
MC005	22.05	347.95	3.1	328.1	8	7.98	280, 65	90, 45
MC006	21.77	341.39	11.0	31.0	7	6.81	280, 65	90, 45
MC011	19.54	344.43	7.7	52.2	8	7.87	259, 23	100, 40
MC012	16.51	357.81	9.5	34.7	8	7.80	259, 23	100, 40
MC013	41.14	347.43	10.7	24.2	9	8.67	259, 23	100, 40
MC014	9.49	338.82	10.2	30.3	8	7.77	186, 26	120, 30
MC015	16.39	337.53	23.9	6.3	7	6.89	186, 26	120, 30
MC030	21.23	316.80	5.1	118.45	8	7.94	338, 26	100, 26
MC033	46.36	335.53	9.2	34.1	8	7.81	228, 53	130, 15
MC034	12.15	337.43	3.1	325.73	8	7.98	345, 37	90, 43
MC035	25.70	304, 99	9.3	36.33	8	7.81	217, 35	58, 25
MC036	1.46	318.15	8.3	45.64	8	7.85	352, 23	123, 22
MC037	5.54	321.04	6.5	72.66	8	7.90	185, 27	102, 18
MC047	10.79	317.46	5.5	100.79	8	7.93	transposed	340, 30

I and D are site mean inclination and declination, respectively. N, R, k and α_{95} are statistical parameters of Fisher [28]. Convention on listing of attitudes is that first number is azimuth 90° clockwise from down dip direction, second number is angle of dip.

and $dm = 8.1°$. As shown in Fig. 6, this pole falls adjacent to the North American apparent polar wander path very near poles from the Summerville Formation of the Colorado Plateau and the Canelo Hills Volcanics of southeastern Arizona. Data for Fig. 6 are summarized in Table 3. It is believed that the well clustered in situ characteristic NRM directions from the McCoy Mountains Formation and positioning of the resultant paleomagnetic pole very near the North American apparent polar wander path is not by chance. Instead, we believe these data provide important age constraints on the age of deformation of the McCoy Mountains Formation.

TABLE 2

Formation mean characteristic NRM results

I	20.6°
D	335.1°
α_{95}	7.7°
k	22.2
N	18
R	17.2
λ_p	57.7°N
ϕ_p	116.2°E
λ_s	33.5°N
ϕ_s	245.5°E

I, D are mean inclination, declination, N is number of sites, R, k, α_{95} are statistical parameters of Fisher [28]. λ_s, ϕ_s (λ_p, ϕ_p) are latitude, longitude of collection locality (pole position).

4. Discussion

The results of the fold test clearly indicate that these sedimentary rocks were magnetized after the development of their relative structures and sympathetic cleavage fabric. Study of opaque mineralogy confirms that the specular hematite resulted from post-depositional precipitation. This precipitation could likely have been in response to the metamorphism associated with the deformation that produced the structural tilting and cleavage.

TABLE 3

Paleomagnetic poles of Fig. 6

Symbol	Name	Age (Ma)	Pole				Reference
			lat. (°N)	long. (°E)	A_{95}(°)	*dp, dm*(°)	
200	200 Ma average	200	68.0	93.0	6.0		[40]
190	190 Ma average	190	72.0	95.0	6.0		[40]
S	Summerville	155–150	67.6	110.8		3.2, 5.4	[37]
MC	McCoy Mountains	165–150	57.7	116.2		4.2, 8.1	this work
CH	Canelo Hills	150	62.2	130.3		4.1, 7.4	[39]
lM	lower Morrison	143–138	61.4	142.2		4.0, 6.5	[41]
uM	upper Morrison		67.5	161.7		3.5, 5.0	[41]
K	Cretaceous average	135–80	67.9	185.9	2.3		[40]

We interpret the placement of the pole position from the McCoy Mountains Formation so close to the Summerville Formation and Canelo Hills Volcanics poles to indicate that magnetization occurred at about the same time as deposition of the Summerville Formation and Canelo Hills Volcanics. The magnetization of the McCoy Mountains Formation is post-depositional, and therefore the paleomagnetic data can only provide a minimum age constraint for the time of deposition of the McCoy Mountains Formation. Other possible interpretations have been pursued, but proved to be either contrary to available geologic constraints or hopelessly ad hoc [13].

The Summerville Formation is dated as probably middle Callovian (late Middle Jurassic) [33,34]. According to the Jurassic time scale of Van Hinte [35], this represents an absolute age of about 150–155 Ma; according to Armstrong [36], it represents an interval of time 165–162 Ma. The Summerville Formation is unfossiliferous in its exposures on the San Rafael Swell, the site of Steiner's [37] paleomagnetic studies, and its geologic age is assigned by inference drawn from regional stratigraphic correlation [38]. The Canelo Hills Volcanics have yielded an age of 151 ± 2 Ma (Rb-Sr isochron [39]), but cannot be assigned to a stage within the Jurassic due to the lack of fossil age control. Based on these stratigraphic considerations, a late Middle to middle Late Jurassic age of magnetization of the McCoy Mountains Formation is proposed. An absolute age of this metamorphic event probably cannot be confined any better than between 165 and 150 Ma. The metamorphism of the McCoy Mountains Formation is certainly older than deposition of the Morrison Formation, of Kimmeridgian age [38] (Fig. 6).

The consistency of the direction of magnetization of the McCoy Mountains Formation from both north- and south-dipping cleavage domains has important implications for the deformation history of the McCoy Mountains Formation. Fabric data collected throughout the McCoy Mountains Formation and underlying Jurassic volcanic sequence as well as in the upper plates of the two thrust faults which bound the McCoy Mountains Formation and Jurassic volcanic sequence imply that the cleavage, tilting, and folding of the upper and lower plates are sympathetic to the bounding thrust faults. This correlation of structures within the McCoy Mountains Formation to the northern and southern bounding thrust faults suggest that the major deformation of the McCoy Mountains Formation is late Middle to middle Late Jurassic.

It must be emphasized that in addition to the major deformation of Jurassic age, evidence of later deformation is common. The later deformation(s) appear to have affected both lower and upper plates to the thrust faults but have resulted in only *minor* disturbances of the earlier fabric. No evidence of major faulting associated with later deformation of the lower or upper thrust plates has been found.

5. Conclusions

The paleomagnetic data, constrained by geologic evidence, indicate that the McCoy Mountains Formation was deposited, deformed and then magnetized in response to a mild metamorphism. The resulting paleomagnetic pole calculated from the observed mean direction of magnetization is located adjacent to the North American apparent polar wander path very near paleomagnetic poles from the Summerville Formation and the Canelo Hills Volcanics. These data indicate that the age of magnetization is late Middle to middle Late Jurassic. Since the age of magnetization is clearly post-deformational, this is a minimum age for deposition of the McCoy Mountains Formation. The age of deposition of the McCoy Mountains Formation is thus bracketed by the loosely dated lower(?) Jurassic volcanic sequence below it and the late Middle/middle Late Jurassic metamorphism which deformed it.

In addition, the consistency of NRM directions of magnetization from both north and south cleavage domains and the sympathetic relationship of these cleavages with the thrust faults which bound the McCoy Mountains Formation suggest that the deformation of the McCoy Mountains Formation was achieved during a single event; that the McCoy Mountains Formation and underlying Jurassic volcanic rocks were squeezed between the North American craton and the composite San Gabriel–Joshua Tree terrane during Jurassic time.

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